

⌊ Rigid Motion

Let V be an inner product space over $\mathbb{R}$.

Def. A map $f: V \rightarrow V$ is called a Rigid Motion if: for all $x, y \in V$,

$$\|f(x) - f(y)\| = \|x - y\|$$

Thm. Let $f: V \rightarrow V$ be a Rigid Motion. Then,

there exists unique orthogonal operator $T: V \rightarrow V$ and translation $g: V \rightarrow V$ s.t. $f = g \circ T$.

Proof. Define

$$T: V \rightarrow V; x \mapsto f(x) - f(o) \quad \text{and} \quad g: V \rightarrow V; y \mapsto y + f(o)$$

$$\text{Then, } g \circ T(x) = g(T(x)) = g(f(x) - f(o)) = f(o) - f(o) + f(o) = f(x). \quad \therefore g \circ T = f.$$

1). T preserves a Norm.

Let $x \in V$ be given.

$$\|T(x)\| = \|f(x) - f(o)\| = \|x - o\| = \|x\|.$$

2). T preserves inner product.

Let $x, y \in V$ be given.

$$\begin{aligned} \|x - y\|^2 &\stackrel{T \text{ is Rigid}}{=} \|T(x) - T(y)\|^2 = \|T(x)\|^2 + \|T(y)\|^2 - 2\langle T(x), T(y) \rangle \\ &= \|x\|^2 + \|y\|^2 - 2\langle T(x), T(y) \rangle \\ &= \|x\|^2 + \|y\|^2 - 2\langle x, y \rangle \end{aligned}$$

$$\text{Thus } \langle T(x), T(y) \rangle = \langle x, y \rangle$$

3). T is linear.

Let $x, y \in V$ and $\alpha \in \mathbb{R}$ be given.

$$\begin{aligned} \|T(\alpha x + y) - (\alpha T(x) + T(y))\|^2 &= \|T(\alpha x + y) - T(y)\|^2 + \|\alpha T(x)\|^2 - 2\langle T(\alpha x + y) - T(y), \alpha T(x) \rangle \\ &= \|\alpha x + y - y\|^2 + \alpha^2 \|x\|^2 - 2\alpha (\langle T(\alpha x + y), T(x) \rangle - \langle T(y), T(x) \rangle) \\ &= 2\alpha^2 \|x\|^2 - 2\alpha (\langle \alpha x + y, x \rangle - \langle x, y \rangle) \\ &= 2\alpha^2 \|x\|^2 - 2\alpha^2 \|x\|^2 = 0. \end{aligned}$$

4). g, T unique.

If $f = T + a = U + b$ for some linear T, U and scalar a, b ,
then $f(o) = a = b$. Thus $T = U$.